

PROJECT

6.4

LOAD DATA INTO DYNAMODB

DATABASE SERIES

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Objective

Learn to create and manage highly scalable NoSQL tables using Amazon DynamoDB. Gain hands-on experience uploading and editing data securely through AWS CloudShell, enabling fast and flexible data handling.

This project helps you understand the power of serverless databases and how DynamoDB scales automatically to handle any amount of data with minimal management.

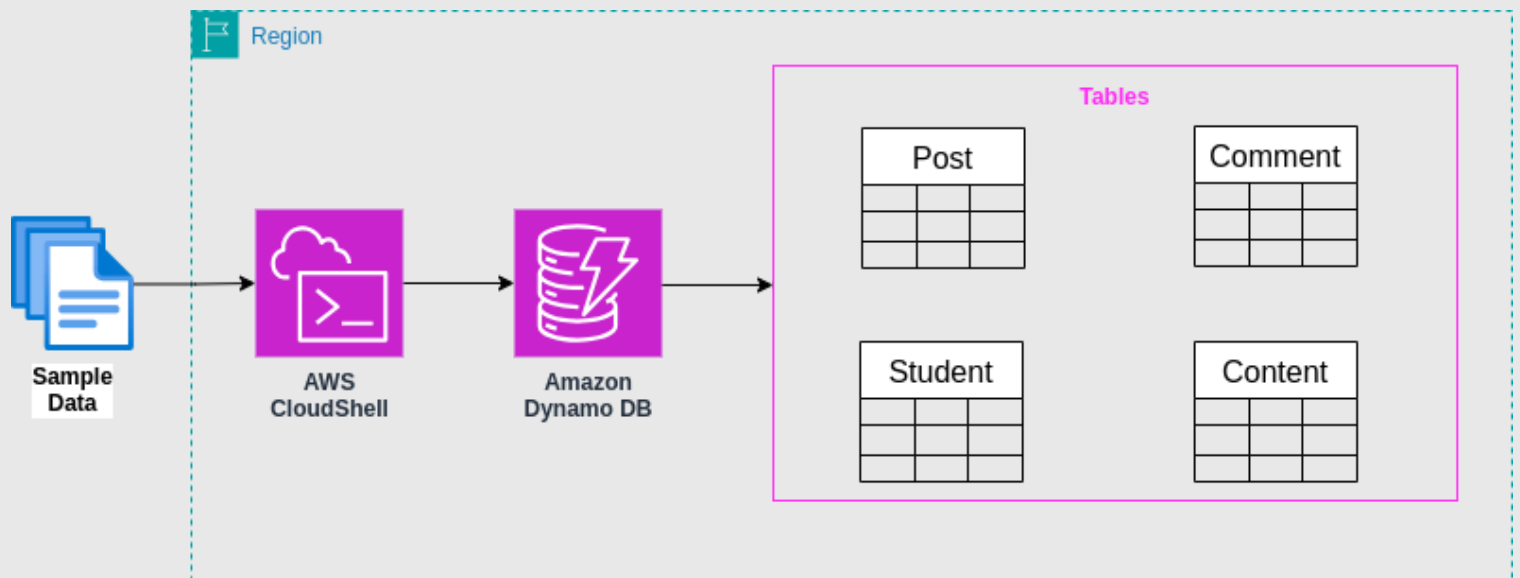
Services

Amazon DynamoDB - For creating and managing scalable NoSQL tables.

AWS CloudShell - For running commands directly in the cloud to interact with DynamoDB.

AWS IAM - To securely manage user permissions and control access to DynamoDB resources.

Architecture Diagram



This project uses Amazon DynamoDB as a fully managed NoSQL database inside your AWS environment. AWS CloudShell acts as your command center, a browser-based terminal with permissions via IAM that lets you create tables, insert, and update data without setting up local tools. All interactions happen securely within the AWS cloud, leveraging DynamoDB's serverless, auto scaling capabilities.

Steps to Implement 1

1. Login with IAM User:

Start by signing in to the AWS Management Console using your IAM user. Ensure the IAM user has permissions for DynamoDB and CloudShell. This keeps your actions secure and avoids using the root account.

2. Create Your First DynamoDB Table:

Head to the DynamoDB console and click “Create table.” Name your table (like MoviesDB) and set the partition key (e.g., MovieID). Leave other settings default for now and create your first NoSQL table.

3. Open AWS CloudShell:

Launch AWS CloudShell from the console’s top bar it’s your terminal in the cloud! This pre configured shell lets you run AWS CLI commands instantly without installing anything locally.

4. Create DynamoDB Tables Using CloudShell:

Inside CloudShell, use the AWS CLI command to create a DynamoDB table. For example, use `aws dynamodb create-table` with proper attributes and key schema. This shows how automation helps in managing resources quickly.

Steps to Implement 2

5. Load Data into Your Tables:

Use CloudShell to insert data into your table using the `aws dynamodb put-item` command. You can also batch upload data by creating a `.json` file and running a script. This gives you full control of data loading.

6. Edit Data in Your Tables:

You can modify or delete items using `update-item` or `delete-item` commands via CloudShell. Test a few updates and explore how the data changes reflect instantly in the DynamoDB console.

7. Monitor and Explore:

Use the DynamoDB console to check table metrics like read/write capacity and data throughput. Try scanning or querying your table to understand how NoSQL searches differ from SQL databases.

8. Clean Up Resources:

Once done, delete the table from the DynamoDB console to avoid any future charges. Also, close CloudShell sessions and review IAM permissions if any were created just for testing.

Story Time

DynamoDB (Your Magic Bookshelf):

Think of DynamoDB as your enchanted bookshelf that stores magical scrolls (data). Each scroll is super fast to find, thanks to magical keys (partition keys). It's serverless and always ready to serve you instantly!

CloudShell (Your Spell Console):

Now meet CloudShell your command line wand that lets you summon AWS services without any setup. Just one click, and you're in a spell room where you control DynamoDB using powerful CLI incantations.

IAM (Your Security Wizard):

Before you begin casting spells, you need a wizard badge that's your IAM user! IAM grants you permission to create tables, load scrolls, and change entries in your magic bookshelf safely and securely.

Data Upload (The Scroll Drop):

Now it's time to add scrolls to your bookshelf! You either write one at a time or drop a whole batch using a magical .json scroll. Your CloudShell wand helps you load them with just a few commands.

Edit & Explore (Magic in Action):

Want to rewrite a scroll or remove it? Just whisper the right CLI spells (update-item or delete-item) and watch the data shift. With DynamoDB, your magic bookshelf always listens and updates in a flash!

BY
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THANK YOU!

